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Heavy metal and antibiotic resistance in four Indian and UK rivers with different levels and types of water pollution

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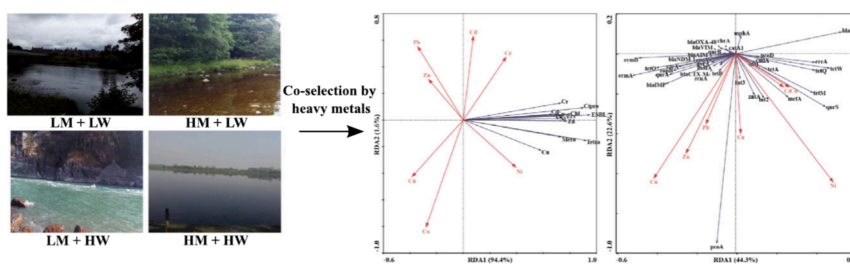
HIGHLIGHTS

- Heavy metal exposure triggers co-selection of heavy metal and antibiotic resistance.
- Significant relationships were found between heavy metal and antibiotic resistance.
- *Int1* could facilitate the co-transmission of heavy metal and antibiotic resistance.
- *rcnA* might play a critical role in antibiotic resistance prevalence in environment.

GRAPHICAL ABSTRACT



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ABSTRACT

Heavy metal pollution can enhance the level of antibiotic resistance, posing concerns to ecosystem and public health. Here, we investigated heavy metal concentrations, heavy metal resistant bacteria and antibiotic resistant bacteria and their corresponding resistant genes, and integrons in four different river environments, i.e., low heavy metals and low wastewater, high heavy metals and low wastewater, low heavy metals and high wastewater, and high heavy metals and high wastewater levels. Heavy metals were found to show positive and significant correlations with heavy metal resistance and antibiotic resistance and integrons ($r > 0.60$, $p < 0.05$), indicating that heavy metal selective pressure can cause heavy metal and antibiotic resistance to be transmitted simultaneously via integrons, which can result in the development of multi-resistant bacteria in the heavy metal-polluted environments. Moreover, there were significant associations between heavy metal resistance and antibiotic resistance ($r > 0.60$, $p < 0.05$), demonstrating heavy metal and antibiotic resistance are connected via a same or related mechanism. Class 1 integrons were found to have strong correlations with heavy metals and heavy metal resistance and antibiotic resistance ($r > 0.60$, $p < 0.05$), indicating a higher occurrence of antibiotic resistance co-selection in the heavy metal-polluted environments.

1. Introduction

Increasing urbanization, industrialization, and population densities have contaminated soils, sediments, and water bodies with heavy metals (and other pollutants), impacting ecosystem functions and human health. Agricultural wastes, mining and industrial releases, urban runoff, treated and untreated domestic wastes, and livestock farming, are the sources of anthropogenic heavy metals to the environment (Laffite et al., 2016; Seiler and Berendonk, 2012; Wei et al., 2019). However, at low levels, many heavy metals are essential for key enzymatic and biochemical activities in animals and microorganisms, although the same heavy metals can be

Abbreviations: HMRB, Heavy metal resistant bacteria; HMRG, Heavy metal resistance gene; MGEs, Mobile genetic elements; ARB, Antibiotic resistant bacteria; ARG, Antibiotic resistance gene; TC, Total coliform; FC, Faecal coliform; ESBL, Extended Spectrum Beta lactam; DO, Dissolved oxygen; EC, Electrical conductivity; TOC, Total organic carbon; TP, Total phosphorus.

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harmful at higher concentrations, resulting in DNA and cellular membranes damage (Knapp et al., 2011).

Heavy metals, being toxic, can influence microbial growth, morphology, and biochemical activities that can result in changes in biomass as well as diversity and community structure (Zampieri et al., 2016; Zhao et al., 2019). Therefore, microbial cells from all environments tightly regulate their cytoplasmic heavy metal concentrations, using different defense mechanisms, including efflux pumps, enzymatic conversion to less toxic forms, reduced heavy metal uptake, heavy metal sequestration, or other options that allow organisms to resist or tolerate heavy metal-induced stress (Bruins et al., 2000; Nies, 2003). Recently, antibiotic resistance has become one of the major threats to global public health. The overuse and misuse of antibiotics for human medicine and agricultural and veterinary practices are the main drivers in the proliferation of antibiotic resistance in the environment. Beside antibiotics, heavy metals also have the tendency to promote the emergence of antibiotic resistance (Gupta et al., 2022). It has been found that heavy metals defense mechanisms are frequently like mechanisms for antibiotic resistance, and therefore, bacterial cells can acquire antibiotic resistance in association with heavy metal resistance under elevated heavy metal exposures (Baker-Austin et al., 2006). Such co-occurrences of heavy metal resistance and antibiotic resistance have been seen in diverse environments, such as rivers, wastewater treatment plants, and associated with agricultural, industrial, and clinical activities (Berg et al., 2005; Di Cesare et al., 2016; Graham et al., 2011; Nakahara et al., 1977; Stepanauskas et al., 2005). Many reports have further indicated that resistance to heavy metal and antibiotic are directly related to heavy metal pollution (Berg et al., 2005; Stepanauskas et al., 2005).

Besides such direct effects, heavy metals also can cause resistance to antibiotics by promoting co-resistance (heavy metal resistance genes (HMRGs) and antibiotic resistance genes (ARGs) exist on the same genetic elements); cross-resistance (the same genetic elements that confers resistance to both heavy metal and antibiotic); and co-regulation mechanisms (regulation of ARGs by heavy metal ions due to associated regulatory processes) (Baker-Austin et al., 2006). Moreover, environmental heavy metal concentrations can be many times greater than antibiotics because they do not readily biodegrade and, if they are bioavailable, can exert long term selection pressure on microbial communities (Stepanauskas et al., 2005). Also, class 1 integrons have been found to offer a selective advantage to bacteria in heavy metal-polluted environments, highlighting the competitive gain of bacteria carrying mobile genetic elements (MGEs) (Rosewarne et al., 2010). In particular, ARGs and HMRGs often co-relate on class 1 integrons (Di Cesare et al., 2016).

Although the potential for antibiotic resistance co-selection exists (Chen et al., 2019), very little data are available that includes both genetic and culturing data for the natural environment. We hypothesized that heavy metals would enhance the occurrence and proliferation of heavy metal resistance as well as antibiotic resistance and thereby favouring co-selection mechanisms in the high heavy metal polluted environments. Additionally, wastewater also contribute a significant proportion of contaminants such as antibiotics, ARB, and ARGs, in addition to heavy metals into the environment, which might have a strong influence on the spread of antibiotic resistance in the environment (Buriánková et al., 2021; Kumar and Pal, 2018). Here we sampled and compared four different river environments with low heavy metals and low wastewater impacts; high heavy metals and low wastewater; low heavy metals and high wastewater; and high heavy metals and high wastewater, including sites in the UK and India. Respective resistant bacteria, HMRGs, ARGs, and MGEs (*int1*, *int2*, and *int3*) were quantified as well as heavy metals and other water quality parameters across systems. Co-occurrence analysis then was used to investigate the relationships between heavy metals, resistant bacteria and genes, and integrons.

2. Materials and methods

2.1. Study sites

The study included three locations on the Tyne River and two on the Ouseburn River in the UK. In India, three locations were on the Ganga River and one location on the River Yamuna (Fig. 1) (see Supporting Information (SI) Table S1 for locations). The Tyne River has comparatively high levels of heavy metals owing to past mining and industrial activities (Macklin et al., 1994). Sites Featherstone (T2) and West Allen (T3) have particularly high levels of Zn, Pb, and Cd (i.e., high heavy metals, low

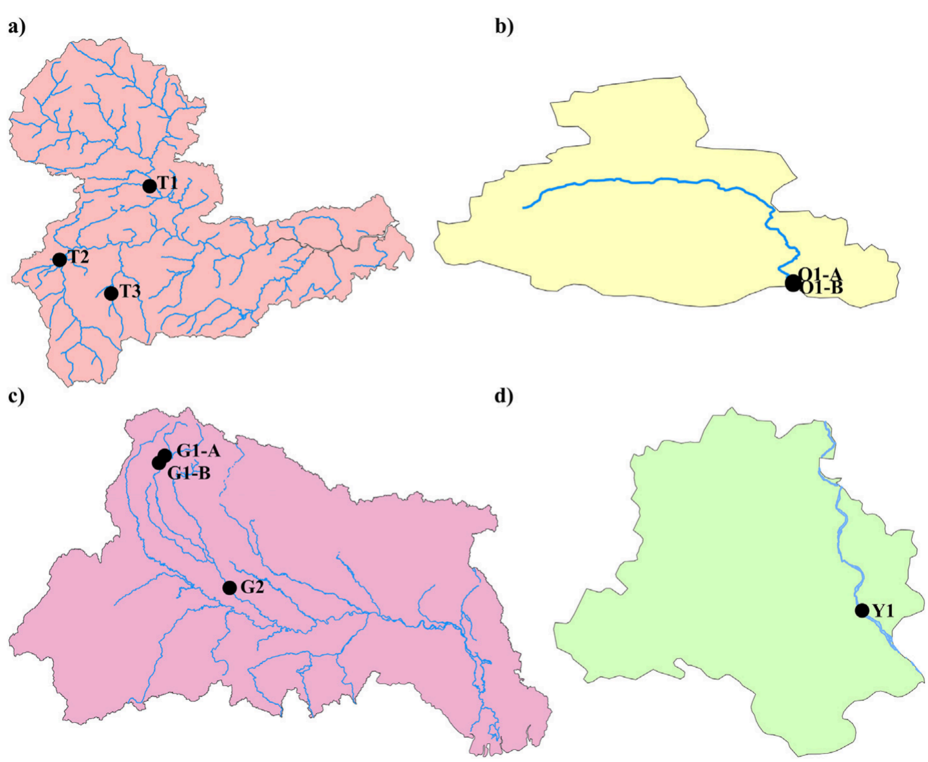

Fig. 1. Location map of the study area. (a) Tyne River (b) Ouseburn River (c) Ganga River and (d) Yamuna river (in Delhi).

wastewater locations); whereas, Warks Burn (T1) is relatively pristine with low heavy metals and low wastewater impacts.

The two Ouseburn sites, upstream (O1-A) and downstream (O1-B), and sites Rishikesh (G1-A) and Haridwar (G1-B) on the Ganga River receive untreated domestic wastewater (Ouseburn is a Combined Sewer Overflow outlet, CSO), and thus are rivers with low heavy metals and high wastewater impacts. Finally, Ganga site Kanpur (G2) is highly polluted with heavy metals from tanneries, textile mills, chemical plants, slaughterhouses, and industrial units as well as untreated domestic wastewater, which also is the case with Nizamuddin (Y1) on the Yamuna River, which is highly polluted with industrial and communities' wastewater. Therefore, G2 and Y1 are high heavy metals and high wastewater sites.

2.2. Sample collection

Surface water and sediments were obtained in triplicate from all locations. All samples were obtained in sterile 500 ml polypropylene containers and taken to the laboratory on ice packs for further analyses. The subsamples for physico-chemical analysis and heavy metal analysis were temporarily stored at 4 °C, while samples for molecular analysis were preserved at −20 °C. The microbial plating work was done within 24 h of sample's collection.

2.3. Physicochemical analysis

pH, temperature (Temp.), electrical conductivity (EC), and dissolved oxygen (DO) of water were estimated in situ with hand-held probes (Hanna Instruments, USA). The sediments pH and EC were assessed with a probe at a sediment and water ratio of 1:5 (g/ml) (Zhang et al., 2019). Ammonium (NH_4^+), nitrate (NO_3^-), and phosphate (PO_4^{2-}) contents in water samples were determined using colorimetric method. In addition, NH_4^+ and NO_3^- were extracted from sediment samples using 2 M KCl and quantified using colorimetric method. Total organic carbon (TOC) concentrations in water and dried sediments were estimated using TOC analyzer (Shimadzu SSM-5000A). Total phosphorus (TP) concentrations in sediments were measured using ICP-MS after digested using HNO_3-HCl (3 : 1, v/v) at 200 °C. The data are presented in Tables S2–S4.

2.4. Heavy metals analysis

Heavy metal concentration was measured using an Agilent 7900 ICP-MS (Agilent Technologies, USA). Heavy metals such as Cd, Co, Zn, Pb, Ni, Cu, and Cr were chosen because they frequently are released into the environment and also have been previously associated with resistance to antibiotics (Graham et al., 2011). Samples were prepared by filtering the water column samples through 0.22μm nylon membrane filters. The sediment samples were acid digested by aqua regia for 30 min at 60 bar and 200 °C with microwave digester. Heavy metal concentrations from the study sites are given in Tables S5–S6.

2.5. Microbial analysis

Total coliform (TC) and faecal coliform (FC) concentrations were estimated using Rapid HiColiform Agar and M-FC Agar, respectively. Heavy metal resistant bacteria (HMRB) and Antibiotic resistant bacteria (ARB) were quantified with Reasoner's 2A agar (R2A) media plates supplemented with a heavy metal or antibiotic at a concentration equivalent to its minimum inhibitory concentration (MIC). The lowest concentration of a heavy metal or antibiotic at which bacterial growth is completely inhibited was defined as its MIC. The MICs used for the heavy metals were Cd - 5 μg/ml, Co - 8 μg/ml, Zn - 20 μg/ml, Pb - 100 μg/ml, Ni - 20 μg/ml, Cu - 10 μg/ml, and Cr - 5 μg/ml (Gupta et al., 2022). The MICs for the antibiotics were chosen from CLSI (2015) or EUCAST (2015) guidelines; i.e., tetracycline – 4 μg/ml, chloramphenicol – 8 μg/ml, ciprofloxacin – 1 μg/ml, erythromycin – 1 μg/ml, and meropenem – 2 μg/ml. HiCrome™ ESBL Agar Supplement (Himedia, India) was used to measure Extended Spectrum Beta lactam (ESBL) resistant bacteria. The total heterotrophs were estimated with R2A Agar media plates without heavy metal or antibiotic.

Microorganisms from sediment samples were “released” into the sterile buffer solution by adding the sediments to sodium pyrophosphate solution (0.2 % (w/v)) in a 1:1 ratio (v/v), which mixed on a shaker table overnight at 150 rpm and 4 °C. Water samples and sediments were serially diluted using sterile phosphate buffer and cultured on R2A Agar supplemented with or without heavy metals or antibiotics and TC and FC plates. Growth conditions were as follows: R2A Agar plates were kept for incubation at 37 °C for three days, and TC and FC plates were retained at 37 °C and 45 °C for 24 h, respectively. Following incubation, the bacterial colonies were counted, and colony-forming units (CFU/ml or CFU/g) were calculated.

2.6. DNA extraction and qPCR

Water samples were filtered through 0.22 μm cellulose nitrate filters. The DNA from the water samples concentrate and sediments was extracted with the Fast DNA Spin Kit for Soil (MP Biomedical, USA) and was serially diluted to reduce the inhibitors effect on qPCR analysis. Based on dilution testing, a 1:100 dilution of DNA was used to quantify the abundance of faecal indicator gene (*BtH*), HMRGs, ARGs, and integrons (*int1*, *int2*, and *int3*). The specific HMRGs and ARGs, chosen for quantification, were based on previous reports assessing heavy metals and antibiotics resistance (Table S7) (Guo et al., 2019; Yin et al., 2017). To estimate the relative faecal contribution in the samples, *BtH* was estimated by targeting the α-1-6 mannanase gene of *Bacteroides thetaiotaomicron* (a faecal-associated bacteria). The 16S rRNA gene also was quantified to normalize *BtH*, HMRGs, ARGs, and integrons levels relative to overall bacterial population size.

All samples were analyzed in triplicate with a BioRad CFX C1000 System (BioRad, Hercules, USA). Each reaction was performed with standards of known concentrations and negative controls. Reaction slope, R^2 values, and efficiencies were within the following range: −3.1 to −3.6, >0.99, and 90 to 110 %, respectively. The primers and reaction programs are given in Table S7.

2.7. Statistical analyses

The analysis was carried out with R software version 3.6.0 (R Core Team, 2019) and Excel 2007 (Microsoft Corporation, USA). The core data were log-transformed prior to the analysis. $p < 0.05$ was used to define statistical significance. As the toxicity of heavy metals depends on their concentrations (Knapp et al., 2011), the cumulative heavy metals concentrations were calculated for each sampling site and sites were classified as low heavy metal or high heavy metal polluted. The data were not normally distributed (Shapiro-Wilk test) and therefore, non-parametric statistical tests were used. Mann-Whitney non-parametric test was conducted to compare bacterial abundance data (e.g., total heterotrophs, TC, FC, HMRB, and ARB concentrations) among sampling sites. Bivariate spearman correlation analysis was done to determine the statistical significance of *BtH*, HMRGs, ARGs, and integrons levels between locations. The genes *rcnB*, *chrT*, *cueO*, and bla_{SHV} were not used for the analysis since these genes were always below quantification limit. Similarly, *BtH* was not detected in sediments and was not included in sediment's analysis. Spearman rank correlation analysis and Redundancy analysis (RDA) were conducted with R 3.6.0 and CANOCO 4.5 (Microcomputer Power, Ithaca, NY), respectively. The correlation among the two items was considered “strong” if r was >0.60 (or < −0.60) and $p <$ 0.05. Bar charts and heatmaps were plotted with R 3.6.0.

3. Results and discussion

3.1. Abundance of total heterotrophs, TC, FC, HMRB, and ARB in different environments

The abundance of culturable total heterotrophs, TC, FC, HMRB, and ARB was quantified across river water and sediment samples representing 1) low heavy metals and low wastewater (LM-LW); 2) high heavy metals and low wastewater (HM-LW); 3) low heavy metals and high wastewater

(LM-HW); and 4) high heavy metals and high wastewater (HM-HW) concentrations. The samples were collected during summer in UK and during winter in India to retain as uniform temperature as feasible, which might have impacted detected resistance markers among locations. However, the water flux, drugs used, and discharge of wastewater in the different seasons might also have impact on the microbial organisms and resistance in waters and sediments. The log abundances are summarized in Figs. 2 and 3. The mean concentrations of total heterotrophs, TC, and FC (Fig. 2) and HMRB and ARB (Fig. 3) were quantified for sites O1-A and O1-B and for G1-A and G1-B. Total heterotrophs, TC, and FC (mean ± standard error) in water samples were 2.30 ± 0.05 to 6.48 ± 0.08; 0.82 ± 0.43 to 5.51 ± 0.04; and Below Detection Limit (BDL) to 4.04 ± 0.17 log CFU/ml, respectively; while in sediment samples were 3.25 ± 0.02 to 7.21 ± 0.14; 1.39 ± 0.08 to 5.30 ± 0.27; and BDL to 4.65 ± 0.09 log CFU/g, respectively. Similarly, HMRB and ARB in water samples were BDL to 6.22 ± 0.04 and BDL to 6.16 ± 0.02 log CFU/ml, respectively, and in sediment samples were 2.34 ± 0.04 to 7.03 ± 0.01 and 1.37 ± 0.17 to 6.65 ± 0.02 log CFU/g, respectively.

The abundances of total heterotrophs, TC, FC, HMRB, and ARB were consistently highest under HM/HW conditions (sites G2 and Y1) followed by LM/HW conditions (sites O1 and G1), especially compared with the LM/LW and HM/LW sites ($p < 0.05$) (Figs. 2 and 3). These patterns broadly suggest that untreated domestic wastewater releases are superficially more important than heavy metal inputs, although the two drivers have mutually amplifying effects as has seen before (Novo et al., 2013). This is logical because domestic wastewater contributes faecal organisms, including resistant strains, and abundant nutrients that can promote microbial growth (Marti et al., 2013). It is noteworthy that although heavy metal concentrations are high at G2 and Y1, in situ microorganisms have clearly adapted to such conditions, which is consistent with much higher levels of HMRB and ARB at those sites (Berg et al., 2005).

In contrast, total heterotrophs, TC, and FC were significantly lower in the water and sediments from the LM/LW (T1) and HM/LW (T2 and T3) sites (Fig. 2), suggesting the lack of domestic wastewater releases result in fewer HMRB and ARBs. Furthermore, the concentrations of few HMRB and ARB were higher at T2 and T3 sites compared to the T1 site (Fig. 3). For example, in the water column, Zn, Cr, Cu, chloramphenicol, and erythromycin-resistant bacteria at T2; and Ni, Pb, and Cr-resistant bacteria at T3 were higher than at T1 ($p < 0.05$). Similarly, in sediments, all heavy metal-, tetracycline, and chloramphenicol-resistant bacteria at T2; and Co, Zn, Ni, Pb, Cu, and ESBL-resistant bacteria at T3 were higher than at T1 ($p < 0.05$). The higher abundances of HMRB and ARB at HM/LW (T2 and T3) sites than LM/LW (T1) site indicates that high heavy metals concentrations, while toxic in nature, may promote both HMRB and ARB in the heavy metal-polluted environments (Stepanauskas et al., 2005).

3.2. Abundance of faecal indicator gene, HMRGs, ARGs, and integrons in different environments

The faecal indicator gene (*BtH*), eleven HMRGs, twenty-six ARGs, and three integrons were measured in river water and sediments with different levels and types of pollutants. The log relative abundances are summarized in Fig. 4. The mean concentrations of *BtH*, HMRGs, ARGs, and integrons were quantified for sites O1-A and O1-B and for G1-A and G1-B. In water samples, log abundances of *BtH* ranged from BDL to (−3.12 ± 0.24) copies/16S rRNA, but in sediments, abundances were BDL. Similarly, log abundances of HMRGs and ARGs in water ranged from BDL to 2.34 ± 0.19 and BDL to 0.72 ± 0.08 copies/16S rRNA, respectively, and for sediment samples, BDL to 2.76 ± 0.24 and BDL to 1.64 ± 0.04 copies/16S rRNA, respectively. The log abundances of integrons in water samples ranged from BDL to (−0.99 ± 0.00) copies/16S rRNA and for sediments, BDL to (−0.23 ± 0.05) copies/16S rRNA. ARGs such as bla_{TEM}, *qnrS*, *tetQ*, and *tetW*; and *int2* were not quantifiable in samples collected from UK rivers due to differential antibiotic use patterns and bacterial hosts in UK and India. The total HMRGs abundances were found higher than total ARGs abundances in all the collected samples. This could be due to higher heavy metals concentration, compared to antibiotics, in the environment as they are not readily degraded and thereby resulting in continuous selective pressure for heavy metal as well as antibiotic resistance (Stepanauskas et al., 2005). In addition, *int1* and *int3* abundances were also found higher than *int2* (Lamba et al., 2018).

3.2.1. Abundance of faecal indicator gene (BtH) in different environments

The abundances of *BtH* were BDL in sediments compared to water, showing that faecal bacteria prefer to dwell in water rather than sediments (Zhou et al., 2019) (Fig. 4). Furthermore, *BtH* abundances were BDL in water samples from LM/LW (T1) and HM/LW (T2 and T3) sites. In contrast, higher levels were found at HM/HW (G2 and Y1) sites, followed by LM/HW (O1 and G1) sites ($p < 0.01$). This indicates that the discharge of greater volumes of faecal waste at HM/HW and LM/HW sites consequentially contributes to accelerate heavy metal and antibiotic resistance at these locations (Seiler and Berendonk, 2012).

3.2.2. Abundance of HMRGs in different environments

Among different HMRGs, *czcA* for resistance to Co, Cd, and Zn; *rcnA* for resistance to Ni and Co; *zntA* for resistance to Pb, Zn, and Cd; *chrA* for

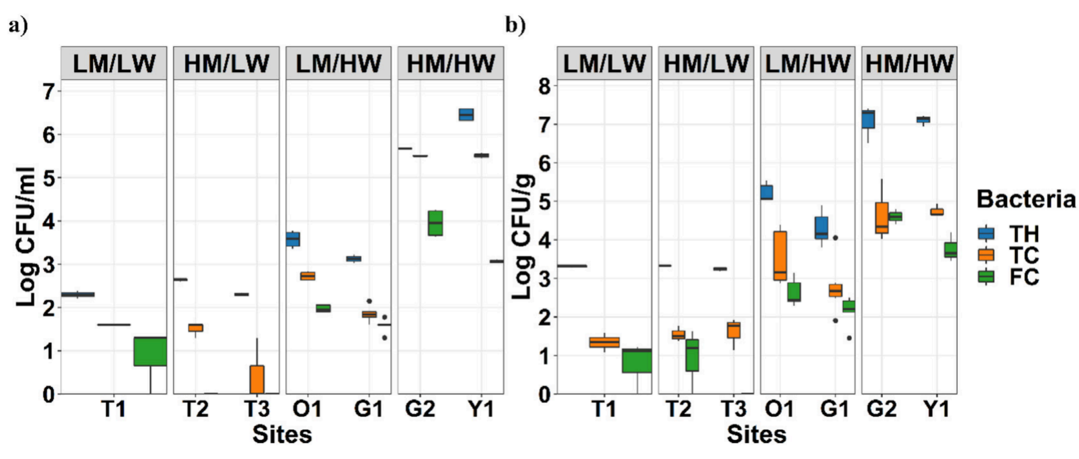

Fig. 2. Abundance of total heterotrophs (TH), TC, and FC in (a) water column and (b) sediments obtained from different environments having 1) low heavy metals and low wastewater (LM/LW); 2) high heavy metals and low wastewater (HM/LW); 3) low heavy metals and high wastewater (LM/HW); and 4) high heavy metals and high wastewater (HM/HW). Box-plot elements represent median, upper and lower quartiles, interquartile range, and outliers. For the water samples, T2 and T3 (FC), and for sediments, T3 (FC), all replicates were below detection limit.

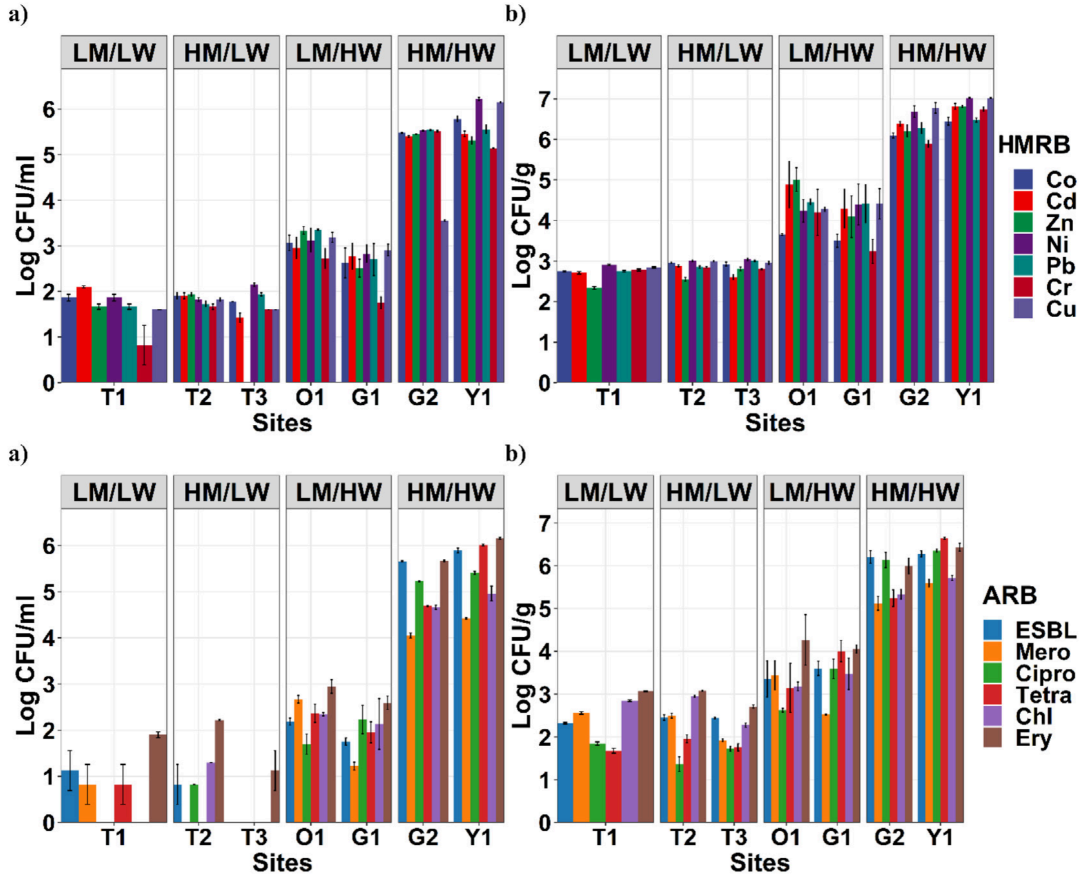

Fig. 3. Abundance of HMRB and ARB in (a) water column and (b) sediments obtained from different environments having 1) low heavy metals and low wastewater (LM/LW); 2) high heavy metals and low wastewater (HM/LW); 3) low heavy metals and high wastewater (LM/HW); and 4) high heavy metals and high wastewater (HM/HW). Columns show the replicates means and vertical bars denote standard error. Tetra, Chl, Cipro, Ery, and Mero represent tetracycline, chloramphenicol, ciprofloxacin, erythromycin, and meropenem-resistant bacteria, respectively. For the water samples T1 (Chl and Cipro-resistant bacteria), T2 (Tetra and Mero-resistant bacteria), and T3 (Zn, ESBL, Mero, Cipro, Tetra, and Chl-resistant bacteria), all replicates were below detection limit.

resistance to Cr; and *cusA* for resistance to Cu were the predominant genes in the samples (Fig. 4). Except for *zntA*, these HMRGs were detected in higher abundances in sediments from the HM/HW site (G2) ($p < 0.01$), implying that high heavy metals concentration in wastewater could be responsible for HMRGs selection and propagation (Reddy and Dubey, 2019). In water samples, however, *czcA*, *rcnA*, and *chrA* were higher at the HM/LW site (T2) ($p < 0.01$), while *cusA* was higher at the LM/LW site (T1) ($p < 0.01$), suggesting that high heavy metals concentrations can results selection of most HMRGs (Chen et al., 2019). The *zntA* was higher in water from HM/HW site (Y1) and sediments from LM/HW site (G1) ($p < 0.01$), indicating that high domestic wastewater contents may have a crucial role in the prevalence of *zntA* in the environment, as the most common host for this gene was *E. coli* (Yang et al., 2020), which is commonly observed in wastewater. In addition, *rcnA* was present in the highest concentration in all the investigated samples. The log abundance of *rcnA* ranged from 1.42 ± 0.07 to 2.34 ± 0.19 copies/16S rRNA in water and 1.85 ± 0.04 to 2.76 ± 0.24 copies/16S rRNA in sediments, showing that multiple copies of *rcnA* were present inside bacterial cells. This may be as a result of its location on both plasmid as well as chromosomal DNA, making it transmissible via both horizontal and vertical gene transfer process. Additionally, *rcnA* was shown associated to ARGs on the chromosomal DNA of an *E. coli* (Gupta et al., 2022), showing *rcnA* may be involved in the emergence of resistance to antibiotics.

3.2.3. Abundance of ARGs in different environments

The $bla_{CTX\text{-}M}$, among ESBL resistant genes, and $bla_{NDM\text{-}1}$ followed by bla_{VIM}, among meropenem resistant genes, were the most common beta-lactam resistant genes (Fig. 4). The $bla_{CTX\text{-}M}$ concentrations were higher in water samples from HM/LW site (T2) and sediments from HM/HW site (G2) ($p < 0.01$), while $bla_{NDM\text{-}1}$ was higher in both water and sediments from HM/LW site (T2) ($p < 0.01$). This may be attributed to high selective pressure exerted from large concentrations of heavy metals, encouraging the process of horizontal gene transmission. In addition, the higher relative abundances of $bla_{CTX\text{-}M}$ in water and $bla_{NDM\text{-}1}$ in water and sediments from site T2 show a higher proportion of bacteria resistant to ESBL and meropenem in water, and meropenem in sediments at site T2. The majority of these ESBL and meropenem resistant isolates also can show resistant to other antibiotics as their resistant genes are frequently located on conjugative plasmids that can also carry resistant genes for others antibiotic (Zurfluh et al., 2013). The bla_{VIM} was higher in water samples from LM/HW site (G1) and sediments from HM/HW site (G2) ($p < 0.01$), suggesting that domestic wastewater discharge has a significant influence on increased ARG abundance (Lamba et al., 2020).

The plasmid mediated *qnrB* and chromosomally associated *gyrA* were the most prevalent ciprofloxacin resistant genes and their abundances were higher in water samples from LM/LW site (T1) and sediments from HM/HW site (G2) ($p < 0.01$) (Fig. 4). The higher *qnrB* and *gyrA* levels at LM/LW and HM/HW sites could be related to high stress conditions caused by limiting nutrients and high DO contents at the T1 site (Table S2) (Son et al., 2018), and the high heavy metals and other pollutants concentrations at G2 site (Xu et al., 2017). The highly stressed environment at T1 and G2 may promote plasmid mediated horizontal transmission of *qnrB* and mutational frequency of chromosomally related *gyrA* in the bacterial community, posing a public health risk. *tetM* and *tetO* were the predominating

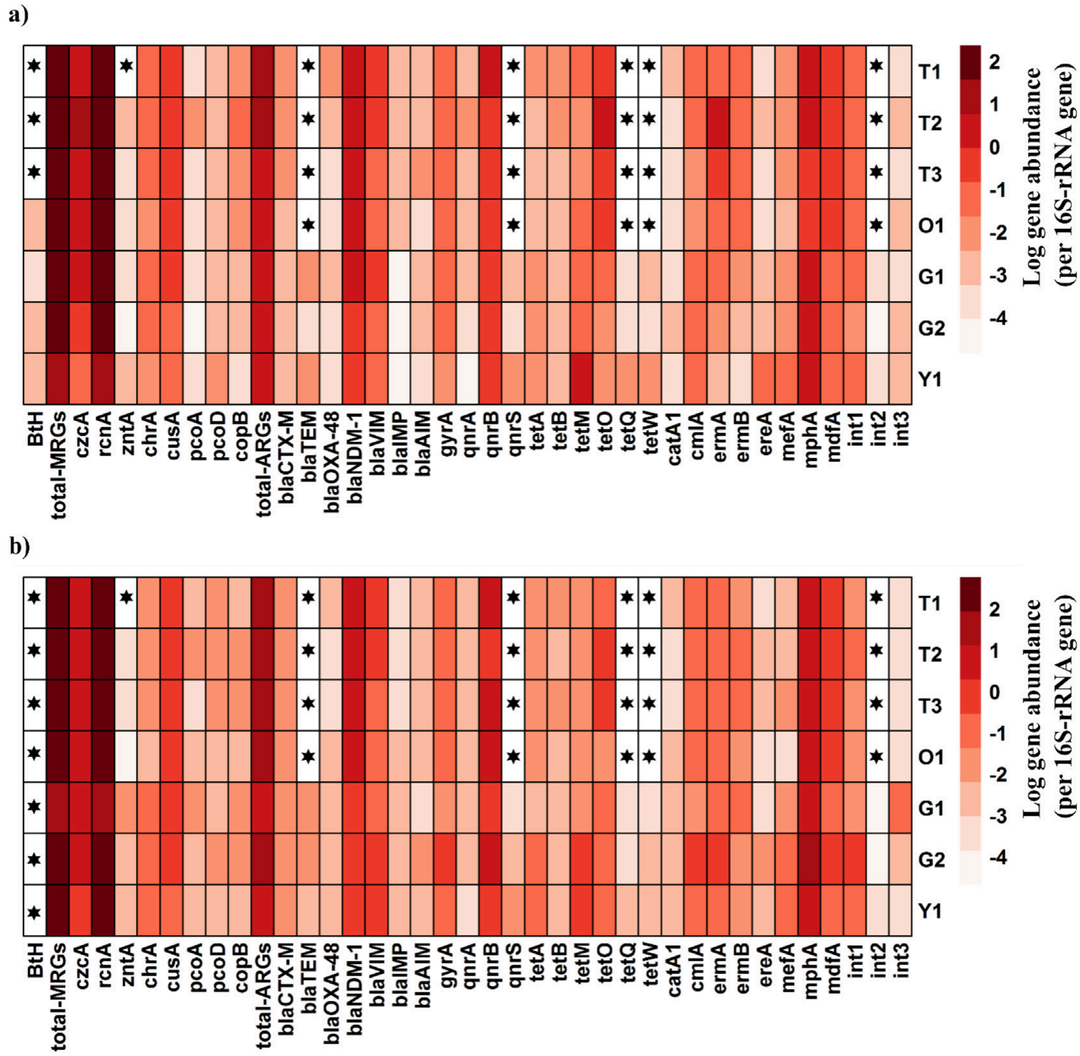

Fig. 4. Relative abundance of faecal indicator gene (*BtH*), HMRGs, ARGs, and integrons in (a) water column and (b) sediments. The black stars indicate samples for which at least one replicate was not quantified. *chrT*, *cueO*, *rcnB*, and bla_{SHV} were not presented because these were below quantification limit.

tetracycline resistant genes in the samples (Fig. 4). The higher prevalence of *tetM* and *tetO* was probably due to their broad host range and several environmental genera often harbored these genes (Zhang et al., 2009). The *tetO* was present in higher abundance in water and sediments from the HM/LW site (T2) ($p < 0.01$), indicating the role of heavy metals in the spread of tetracycline resistance by horizontal gene transmission (Hu et al., 2017). However, a higher concentration of *tetM* was observed in water samples from HM/HW site (Y1) and sediments from HM/HW site (G2) ($p < 0.01$). This could be because *tetM* has a broad host range and the higher bacterial abundance at HM/HW sites favours the prevalence of *tetM* within the bacterial community at these sites.

Furthermore, *cmlA* was the most frequent chloramphenicol resistant gene in the samples, showing that efflux could be the most common mechanism for chloramphenicol resistance (Ma et al., 2019) (Fig. 4). The *cmlA* was present in higher abundance in water samples from LM/LW site (T1) and in sediment samples from HM/HW site (G2) ($p < 0.01$), suggesting higher stress conditions at T1 and G2 could be responsible for *cmlA* dissemination.

Among erythromycin resistant genes, *mphA* followed by *ermA* were the most dominant genes at all sites except at Y1, where *mefA* abundances were higher than *ermA* (Fig. 4). The increased *mefA* abundance at Y1 might be due to shifts in microbial community composition under low DO conditions at this site (Table S2) (Tao et al., 2016). The higher levels of *mphA*, *ermA*, and *mefA* suggest that antibiotic inactivation followed by target site alteration and efflux were the most prominent mechanisms for erythromycin resistance. These resistance genes were present in higher concentrations in sediments from HM/HW site (G2) ($p < 0.01$), while, in water samples, *mphA*, *ermA*, and *mefA* abundances were higher at LM/LW (T1), HM/LW (T2), and HM/HW site (Y1), respectively ($p < 0.01$). The higher concentrations of these genes in sediments from the G2 site implies that high levels of heavy metals and other contaminants cause an increase in erythromycin resistance levels in the environments (Xu et al., 2017).

The multidrug resistant gene, *mdfA*, also was quantified in this study and its abundance was higher in water samples from LM/LW site (T1) and sediments from HM/HW site (G2) ($p < 0.01$) (Fig. 4). The increased *mdfA* levels at these sites show the possibility of multidrug resistant bacteria developing at these locations, raising substantial public health concerns as multidrug resistance is more difficult to treat than resistance to a single antibiotic.

3.2.4. Abundance of integrons in different environments

The proliferating genes *int1*, *int2*, and *int3* also were quantified to estimate the integrons' role in the co-transmission of resistance to antibiotics (Fig. 4). The *int1* was found higher in water samples from HM/HW site (Y1) and in sediments from HM/HW site (G2) ($p < 0.01$), indicating a higher abundance of *int1* in highly heavy metals and wastewater contaminated environments (Gillings et al., 2015; Wright et al., 2008). Additionally, since the relative abundance of most HMRGs and ARGs was higher in sediments from G2, the higher relative abundance of *int1* at G2 demonstrates the possible role of *int1* in the co-transmission of ARGs. Furthermore, *int3* levels were observed higher in water samples from LM/HW site (O1) and sediments from LM/HW site (G1) ($p < 0.01$), indicating high domestic wastewater concentrations might promote *int3* mediated transmission of resistant genes into the environment.

3.3. Correlation between coliforms, faecal indicator gene, resistance to heavy metals and antibiotics, and integrons

Bivariate Spearman correlation analysis was carried out among coliforms (TC and FC); faecal indicator gene (*BtH*); resistant bacteria and genes; and integrons to find out the relationships between wastewater and resistance to heavy metals and antibiotics (Table S8). The concentrations of TC and FC were found to be positively and significantly correlated with all HMRB and ARB in water and sediments ($r \geq 0.75$, $p < 0.05$), showing the important contribution of wastewater in increasing resistance to heavy metals as well as antibiotics in the environments (Lamba et al., 2020; Seiler and Berendonk, 2012). The significant associations between coliforms and HMRB and ARB shows these bacterial groups tend to acquire both heavy metal and antibiotic resistance; therefore, pose a potential health risk. In addition, TC and FC levels also were positively correlated with *tetM* and *int1* in sediments ($r \geq 0.71$, $p < 0.05$), showing that these bacteria are receptive to ARG and can propagate them to other pathogens by horizontal gene transmission (Mishra et al., 2018). Some negative correlations of TC and FC with heavy metal and antibiotic resistance genes ($r < -0.60$, $p < 0.05$) also were observed. This could be because TC, FC, and bacterial hosts carrying HMRGs and ARGs might have originated from different sources or belong to separate groups and thereby showing inhibitory effects. Furthermore, the faecal indicator gene, *BtH*, showed positive correlations with TC, FC, and all HMRB, except Cu resistant bacteria, and ARB in water samples ($r > 0.60$, $p < 0.05$), representing faecal matter as a source of HMRB and ARB (Lamba et al., 2020; Seiler and Berendonk, 2012).

3.4. Correlation between physicochemical parameters, resistance to heavy metals and antibiotics, and integrons

Spearman correlation analysis showed that physicochemical parameters including EC, TOC, NO_3^-, and PO_4^{2-} in water and EC, NH_4^+, and NO_3^- in sediments were positively correlated with most HMRB and ARB ($r > 0.60$, $p < 0.05$) (Fig. S1), directly affecting the HMRB and ARB levels. However, DO contents in water showed negative correlations with Co, Ni, Cu, tetracycline and erythromycin-resistant bacteria (Fig. S1a), indicating high DO levels likely decreased the HMRB and ARB levels. In addition, temperature, DO, TOC, NO_3^-, and PO_4^{2-} in water and pH, EC, TOC, NH_4^+, NO_3^-, and TP in sediments showed positive correlations with HMRGs and ARGs ($r > 0.60$, $p < 0.05$) (Fig. S2), showing that environmental factors may influence the HMRGs and ARGs abundances. Previous study also has indicated a positive correlation between temperature and an ARG, suggesting that high water temperature also facilitated the propagation of ARGs (Wang et al., 2020). Nutrients, such as TOC, NH_4^+, NO_3^-, PO_4^{2-}, and TP could offer substrate for bacterial growth, including HMRB and ARB growth, and higher nutrient levels were better for horizontal gene transmission (Yuan et al., 2018). Moreover, NH_4^+ and NO_3^- in sediments showed positive correlations with *int1* ($r > 0.70$, $p < 0.05$) (Fig. S2b) (Lu et al., 2019), demonstrating that ammonium and nitrate might have a significant role in the transmission of HMRGs and ARGs via *int1*.

3.5. Correlation between heavy metals, resistance to heavy metals and antibiotics, and integrons

Positive and significant correlations of heavy metals with HMRB and ARB were observed in the sediments ($r > 0.60$, $p < 0.05$) (Fig. 5), demonstrating that heavy metals tend to proliferate both HMRB and ARB in the environments (Wright et al., 2006). On the contrary, no significant correlations exist between heavy metals concentrations and resistant bacteria in water samples because of low heavy metals concentrations in water. Furthermore, redundancy analysis (RDA) was conducted demonstrating how the proportion of the HMRB and ARB changes can be accounted by heavy metals in water and sediments (Fig. S3). Heavy metals could account for 96.0 % and 96.5 % of the changes in HMRB and ARB levels in water and sediments, respectively.

Heavy metals also showed positive and significant relations with HMRGs, ARGs, and *int1* abundances in water and sediments ($r > 0.60$, $p < 0.05$) (Chen et al., 2019) (Fig. 6). Additionally, some negative associations also were seen between heavy metals and ARGs ($r < -0.60$; $p < 0.05$), which may be due to the inhibition of specific bacterial populations carrying ARGs by heavy metals (Knapp et al., 2017). RDA analysis also indicated heavy metals can explain 66.9 % and 75.9 % changes in HMRGs, ARGs, and integron levels in water and sediments, respectively (Fig. S4).

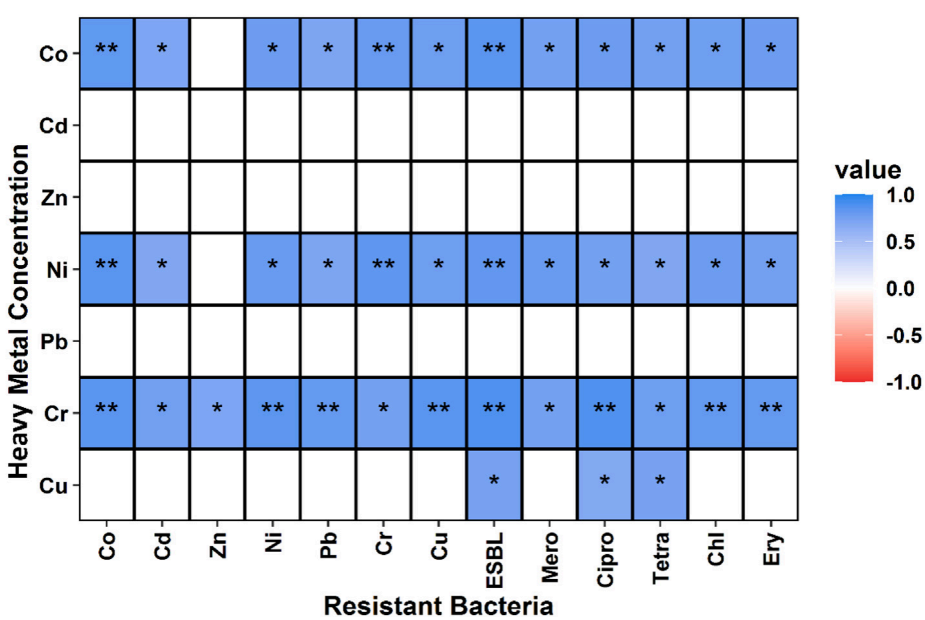

Fig. 5. Correlation between heavy metals and resistant bacteria abundances in sediments ($p < 0.05$). No significant correlations were observed in the water column. The level of significant correlation was shown by asterisk (*, $p < 0.05$; **, $p < 0.01$; and ***, $p < 0.001$). Each block's color represents Spearman's correlation coefficient.

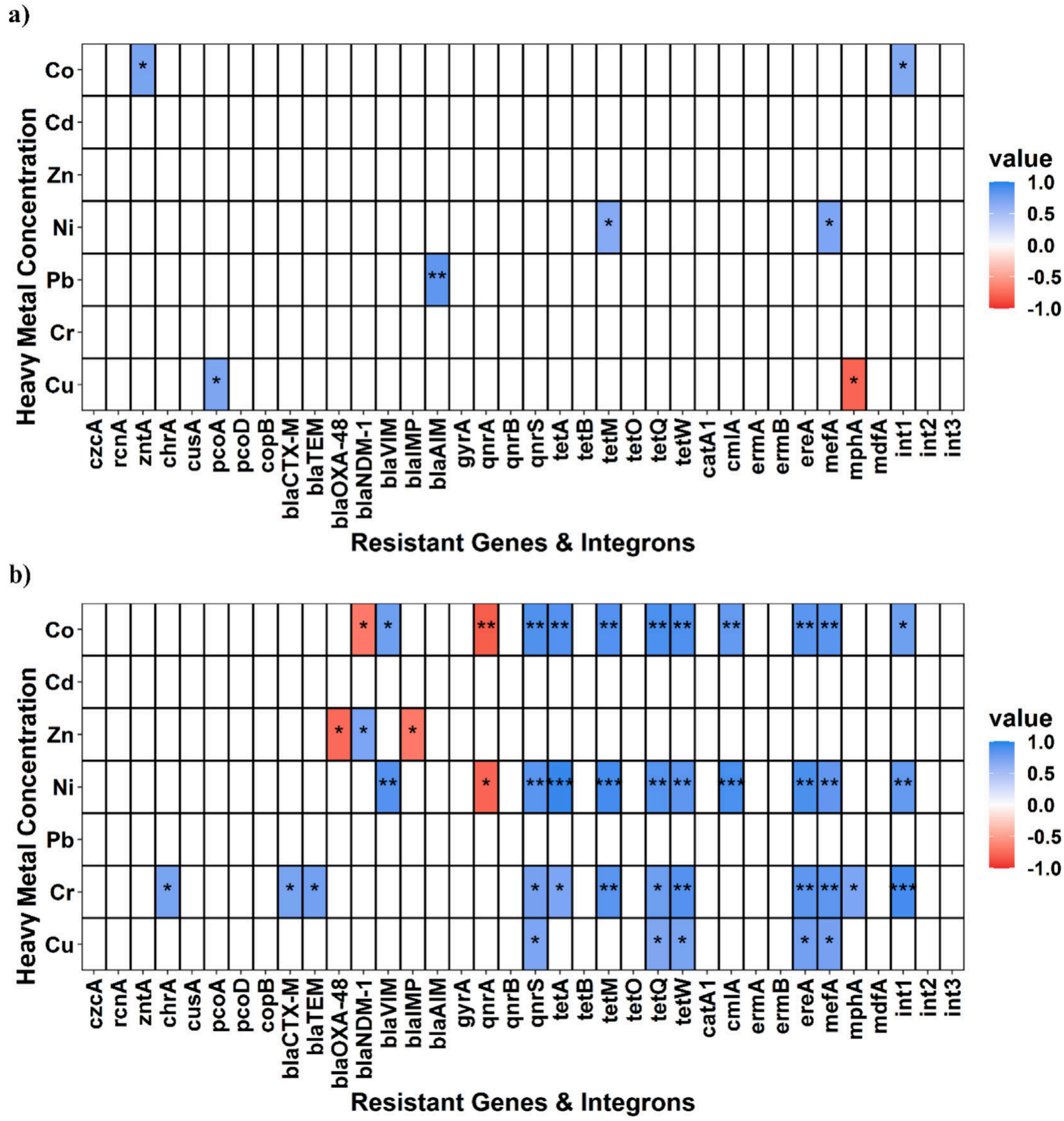

Fig. 6. Correlation between heavy metals and resistant genes and integrons abundance in (a) water and (b) sediments ($p < 0.05$). The level of significant correlation was shown by asterisk (*, $p < 0.05$; **, $p < 0.01$; and ***, $p < 0.001$). Each block's color represents Spearman's correlation coefficient.

The positive relationships of heavy metals with resistance to heavy metals and antibiotics, and integrons indicate that heavy metal selection pressure can induce the transmission of HMRGs and ARGs via integrons that could results in the development of HMRB and ARB in the environment. For example, Cr contamination may facilitate the spread of *chrA* which was found to be located downstream of *int1* and thus could confer co-resistance to chromate and antibiotics (Yang et al., 2020). In our study, Cr concentrations were positively correlated with *chrA*, various ARGs, and *int1* as well as all HMRB and ARB in sediments (Figs. 5 and 6b), demonstrating the possibility of occurrence of Cr induced proliferation of antibiotic resistance in the environments.

Additionally, the selective pressure due to heavy metals can trigger some efflux genes which can simultaneously pump both antibiotic and heavy metal out of the cell and thus, increase the abundances of HMRB and ARB (Levy, 2002). TetA(L) efflux pump can confer resistance to tetracycline and cobalt by pumping out tetracycline and cobalt out of the cell (Cheng et al., 1996). This research also found positive associations between Co concentration and *tetA*, Co- and tetracycline-resistant bacteria in sediments (Figs. 5 and 6b).

Furthermore, heavy metal Zn is essential for the activity of metallo-β-lactamases like bla_{NDM-1}, which catalyze the breakdown of β-lactam antibiotics, causing bacteria to become antibiotic resistant (Nordmann et al., 2012). Zn in sediments was found to positively correlate with bla_{NDM-1} in this study (Fig. 6b). Thus, these findings indicate that heavy metals selective pressure could have a vital role in the emergence and maintenance of antibiotic resistance in the environment (Xu et al., 2017).

3.6. Correlation between resistance to heavy metals and antibiotics and integrons

Spearman correlation analysis was conducted to investigate the associations between heavy metals resistance and antibiotics resistance and integrons ($r > 0.60$ (or < -0.60), $p < 0.05$). The concentration of all HMRB was strongly positively correlated with all ARB in water and sediments ($r > 0.80, p < 0.01$) (Fig. S5), showing the simultaneous occurrence of HMRB and ARB in the environments (Berg et al., 2005). This can be owing to different HMRGs and ARGs may be present on the same genetic element including the same genome or plasmid (He et al., 2017).

Furthermore, HMRB and ARB exhibited positive relationships with ARGs in water and sediments ($r > 0.60, p < 0.05$) (Fig. S6), demonstrating that ARGs can make bacteria resistant to both heavy metals and antibiotics. This can be explained by sharing of resistance mechanisms between heavy metals and antibiotics (Baker-Austin et al., 2006), as well as the existence of

HMRGs and ARGs on the same genetic units (He et al., 2017), leading to co-occurrence of resistance to antibiotics in the environments.

Moreover, all HMRB and ARB exhibited positive relations with *qnrS* (excluding Cd and Zn-resistant bacteria) and *tetM* in sediments (Fig. S6b). This is because these genes are often found on mobile units and can be transmitted across bacteria via horizontal gene transmission, resulting in a wide host range (Le et al., 2015; Zhang et al., 2009). Some negative associations of resistant bacteria with resistant genes also were seen in water and sediments ($r < -0.60$, $p < 0.05$) (Fig. S6), that may be due to competition and inhibition between resistant bacteria and bacteria carrying resistant genes. Moreover, positive associations between resistant bacteria and resistant genes were found to be greater in sediments than water (Fig. S6), implying that sediments provide better conditions (heavy metals, nutrients, high bacterial loads, HMRGs, and ARGs) for resistant gene proliferation and the emergence of multi-resistant bacteria.

HMRGs abundance in water and sediments also was positively related with ARGs abundance ($r > 0.60$, $p < 0.05$), revealing that HMRGs and ARGs may co-occur in the environments (Fig. S7) (Chen et al., 2019). The positive relationships among HMRGs and ARGs are most likely because both are found on the same mobile elements (Baker-Austin et al., 2006). These findings indicate that HMRGs have a critical role in ARGs proliferation by horizontal gene transmission (Guo et al., 2019), posing potential environmental and human health risks.

Furthermore, class 1 integrons in sediments were found to show positive relationships with all HMRB and ARB, as well as HMRG, *chrA*, and various ARGs ($r > 0.60$, $p < 0.05$) (Table S9), indicating that *int1* have a vital role in the transmission of resistance to heavy metals and antibiotics in the environments (Di Cesare et al., 2016). Besides, *int2* also was significantly correlated with a few HMRGs and ARGs in water and sediments. The significant correlations among integrons and HMRGs and ARGs indicate that horizontal gene transfer could be linked to antibiotic resistance co-selection in the environment.

4. Conclusions

This research provides evidence that high concentrations of heavy metals had a great ability to increase antibiotic resistance, resulting in the occurrence of multi-resistant bacteria. The heavy metal-induced antibiotic resistance may increase when high heavy metal concentrations and high wastewater contents are combined. In addition, as heavy metals are more persistent compared to antibiotics, their accumulation in the environments becomes highly serious and threatens public health. Therefore, it is imperative to control heavy metals pollution in the environments to reduce the transmission of heavy metals induced antibiotic resistance co-selection. This study offers a new scientific basis for further understanding the specific phenomenon for co-selection of resistance to antibiotics in environments with varying quantities and types of contaminants.

CRediT authorship contribution statement

Dr. Sonia Gupta: Methodology, Investigation, Visualization, Conceptualization, Writing - Reviewing and Editing, Writing - Original draft preparation. **Dr. Shaikh Ziauddin Ahammad:** Supervision, Conceptualization, Reviewing and editing, Funding acquisition. **Prof. David W. Graham:** Reviewing and editing, Conceptualization. **Prof. T.R. Sreekrishnan:** Reviewing and editing, Supervision.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.scitotenv.2022.159059.

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